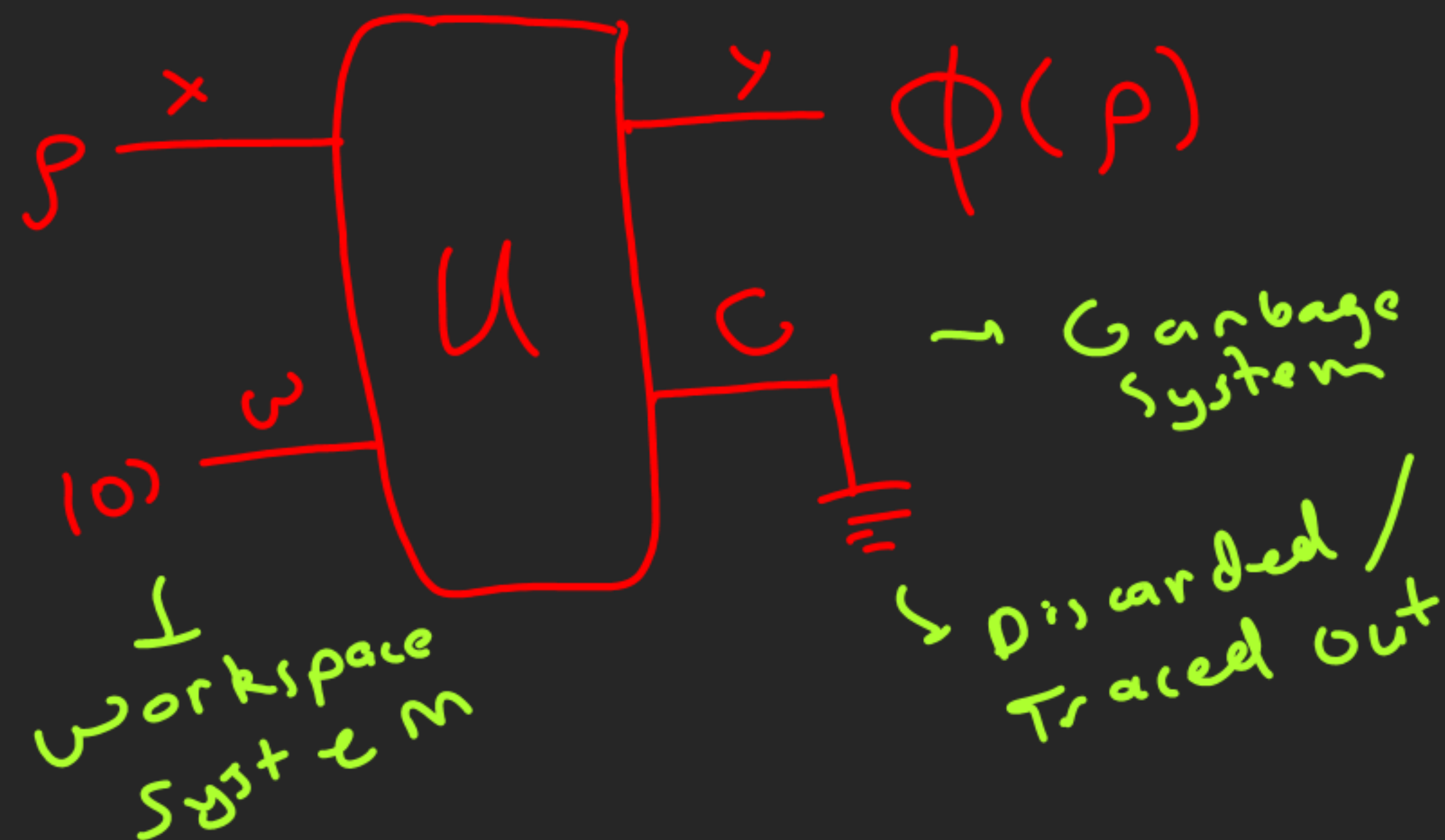


Channel Representations

Stinespring Representation



$$\begin{aligned}\Phi(\rho) &= \text{Tr}_G(\mathcal{U}(\rho \otimes |0\rangle\langle 0|)\mathcal{U}^\dagger) \\ &= \text{Tr}_G(V\rho V^\dagger)\end{aligned}$$

$$V = \mathcal{U}(I_x \otimes |0\rangle)$$

Kraus Representation

$$\Phi(\rho) = \sum_{k=0}^{N-1} A_k \rho A_k^\dagger$$

$$\sum_{k=0}^{N-1} A_k^\dagger A_k = I \rightarrow \text{Trace Preservation}$$

$$\Phi(x) = y$$

$x \rightarrow \text{Input} \rightarrow n$ classical states
 $y \rightarrow \text{Output} \rightarrow m$ classical states

Kraus Matrices $\rightarrow \{A_0, \dots, A_{N-1}\}$
for $N = n \cdot m$, $\{A_0, \dots, A_{N-1}\}$ will always exist

Choi Representations

Channel $\phi \rightarrow$ Represented by $J(\phi)$

Input System $\rightarrow n$ classical states

Output System $\rightarrow m$ classical states

$J(\phi) \rightarrow nm \times nm$ matrix

$\rightarrow$ It's a faithful representation $\rightarrow J(\phi) = J(\psi)$ if and only if $\phi = \psi$

$\rightarrow$ conditions on $J(\phi)$ are true if and only if ϕ is valid

$$\text{Tr}_Y \left(\frac{\mathcal{J}(\Phi)}{n} \right) = \frac{I_X}{n} \quad \text{Tr}_Y(\mathcal{J}(\Phi)) = I_X$$

$$\text{Tr}_Y(\mathcal{J}(\Phi)) = \sum_{a,b \in \Sigma} |a\rangle\langle b| \text{Tr}(\Phi(|a\rangle\langle b|))$$

$$= \sum_{a,b \in \Sigma} |a\rangle\langle b| \text{Tr}(|a\rangle\langle b|)$$

$$= \text{Tr}_X(\mathcal{J}(\Phi)) = \sum_{a \in \Sigma} |a\rangle\langle a| = I_X$$

$$\Phi(\rho) = \text{Tr}_X(\mathcal{J}(\Phi)(\rho^T \otimes I))$$

→ prove

Equivalence of Representations

General Channel $\rightarrow$ Transforms X into Y

$\rightarrow$ Density matrices to density matrices

Choi Representation $\rightarrow \mathcal{J}(\Phi) \geq 0$

$$\rightarrow \text{Tr}_Y(\mathcal{J}(\Phi)) = I_X$$

Kraus Representation $\rightarrow \Phi(\rho) = \sum_{k=0}^{N-1} A_k \rho A_k^\dagger$

$$\rightarrow \sum_{k=0}^{N-1} A_k^\dagger A_k = I$$

Stinespring Representation $\rightarrow (X, \omega) \rightarrow (Y, \phi)$

$$\rightarrow \Phi(\rho) = \text{Tr}_G(u(\rho \otimes |0\rangle\langle 0|)u^\dagger)$$

General channel to Choi Representation

$$\begin{aligned} \mathcal{J}(\phi) &\rightarrow \text{Density Matrix} \rightarrow \mathcal{J}(\phi) \geq 0 \\ &\quad \rightarrow \mathcal{J}(\phi) = (\mathcal{J}(\phi))^T \\ &\quad \perp \end{aligned}$$

Transforms Density Matrix
into Density Matrix

$$\text{Tr}_Y(\mathcal{J}(\phi)) = I_X \quad \rightarrow \quad \mathcal{J}(\phi) \text{ preserves Trace}$$

Choi Representation to Kraus Representation

$$J(\phi) \geq 0 \rightarrow J(\phi) = \sum_{k=0}^{n-1} |\psi_k\rangle\langle\psi_k| \rightarrow \text{Spectral Theorem can be used}$$

$$|\psi_k\rangle = \sqrt{\lambda_k} |Y_k\rangle$$

$$J(\phi) = \sum_{k=0}^{n-1} \lambda_k \underbrace{|Y_k\rangle\langle Y_k|}_{\substack{\downarrow \\ \text{Eigenvectors}}} \underbrace{\lambda_k}_{\substack{\downarrow \\ \text{Eigenvalues}}}$$

$$|\psi_k\rangle = \sum_{a \in \Sigma} |a\rangle \otimes |\phi_{k,a}\rangle$$

$$|\phi_{k,a}\rangle = (\langle a| \otimes I_Y) |\psi_k\rangle$$

$$\text{For } A_k = \sum_{a \in \Sigma} |\phi_{k,a}\rangle\langle a|, \quad \phi(\rho) = \sum_k A_k \rho A_k^\dagger$$

$$\left(\sum_{k=0}^{n-1} A_k^\dagger A_k \right)^T = \text{Tr}_Y \{ J(\phi) \} = I_X \rightarrow \text{Prove}$$

↓

Trace Preservation

Kraus Representation to Stinespring Representation

$$U = \sum_{i,j=0}^{n-1} M_{i,j} \otimes |i\rangle\langle j| \quad \text{where } M_{k,0} = A_k$$

$$\text{Tr}_G (U(\rho \otimes |0\rangle\langle 0|)U^\dagger) = \text{Tr}_G (U\rho U^\dagger)$$

$$V = U(I_X \otimes |0\rangle\langle 0|)$$

$$= V = \sum_{i,j=0}^{N-1} (M_{i,j} \otimes |i\rangle\langle j|) (I \otimes |0\rangle\langle 0|)$$

$$= \sum_{i=0}^{N-1} M_{i,0} \otimes |i\rangle\langle 0| = \sum_{k=0}^{N-1} A_k \otimes |k\rangle\langle 0|$$

$$\text{Tr}_G(V \rho V^\dagger) = \sum_{k=0}^{N-1} A_k \rho A_k^\dagger = \Phi(\rho)$$

→ Gram-Schmidt Orthogonalisation process can be used to select remaining elements in $\mathcal{U}$

Stinespring Representation to General Channel

$\rho \rightarrow$ Density matrix $\rightarrow \Phi(\rho) = \text{Tr}_G(V \rho V^\dagger) \rightarrow$ Density matrix

$\rho \rightarrow \Phi(\rho) \rightarrow$ Trace is preserved